Joint Design of Frequency and Bandwidth for Multifrequency SAR Based on Mutual Information Maximization

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***Abstract*— The carrier frequency and bandwidth are vital parameters of radar transmit signals. In this article, a joint design of frequency and bandwidth in multifrequency synthetic aperture radar (SAR) is proposed to improve the target information acquisition capability. First, the target detection mutual information (MI) is considered as the performance metric, and its mathematical expression is derived using the example of decision-level fusion and hypothesis testing. The design is formulated as an optimization problem with multiple engineering constraints based on MI maximization. Then, a modified genetic algorithm (GA) is proposed to find the optimal solution satisfying the constraints via a code adjustment operator. It is shown by simulation results that for a specific scene, the multifrequency SAR with its frequencies and bandwidths designed by the proposed method has higher target information acquisition capability and more accurate target detection performance than existing spaceborne SAR systems.**

***Index Terms*— Bandwidth, frequency, genetic algorithm (GA), mutual information (MI), synthetic aperture radar (SAR), target detection.**

I. Introduction

SYNTHETIC aperture radar (SAR) imaging has been used in many fields such as agriculture [1], hydrology [2], forestry [3], oceanography [4], geology [5], and defense [6]. Some typical SAR applications, such as target detection, classification, and parameter inversion, depend on the amount of target information contained in the acquired images. Therefore, how to obtain more target information in the SAR imaging process has become one of the key challenges in the area.

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Multifrequency SAR transmits and receives signals with multiple carrier frequencies and therefore can obtain more target information compared with the single-frequency SAR by acquiring images at different frequencies [7], [8]. In addition, a high-resolution SAR image contains more detailed target information such as texture, contour, and geometric structure [9], [10], while the range resolution of SAR is mainly determined by the bandwidth of transmit signals. As a result, the carrier frequency and bandwidth of transmit signals play a crucial role in target information acquisition of SAR, and it is necessary to design the parameters of a multifrequency SAR to achieve the maximum acquisition of target information.

The existing carrier frequency selection methods mainly rely on qualitative analyses based on varying radar applications. The frequency selection problem was investigated in [11] for anti-jamming in radar netting, to avoid jamming and some specific frequency bands. The relationship between the carrier frequency of space-based radar and its parameters, including the size and weight of the antenna and power consumption of the load, was analyzed in [12]. In [13], the target resonant frequency was selected as the carrier frequency of anti-stealth radar to increase the target RCS and achieve anti-stealth. The relationship between gun tubes RCS and the carrier frequency of the interior ballistic measurement radar was provided in [14] with some suggestions on frequency selection to reduce multimode noise. For SAR, the carrier frequency is usually set to some empirical values by analyzing the image quality at different bands [15], such as 1.2 GHz in the L-band, 5.4 GHz in the C-band, and 9.6 GHz in the X-band [16]. However, frequencies which can potentially provide a lot of information for some specific targets may be missed because the empirical frequencies may not be able to maximize the acquired target information. To obtain target information as much as possible, it is necessary to design the carrier frequencies for multifrequency SAR in a quantitative way.

On the other hand, the bandwidth of the transmit signal has been extensively studied because it determines the radar resolution. Since the introduction of linear frequency-modulated (LFM) signal in 1951 [17], most of the research has been devoted to improving the time bandwidth product (TBP) of the signal [18], [19], as it is widely accepted that a large TBP can improve the signal-to-noise ratio (SNR) and further increase the accuracy in target detection and classification [20]. However, most targets exhibit extended characteristics in high

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resolution and the point target models are no longer suitable. It was pointed out in [21] that the detection performance of a wideband radar is worse than that of a narrowband one at low SNR under the scattering center model of the target. Moreover, high resolution makes the interference of the background in the SAR image not negligible. Blindly increasing the bandwidth can be counterproductive for target detection in certain SNR ranges [22], [23]. Although signal bandwidth is an important design parameter for target information acquisition, there is no existing research about SAR bandwidth design to maximize the acquisition of target information in open literature.

Mutual information (MI) between target and radar output signal was first used to represent the target information acquisition capability of radar in [24], where a water-filling algorithm was proposed for transmit waveform design based on MI maximization. The water-filling method was further optimized by considering clutter [25]. The SAR waveform design based on MI maximization with high range resolution constraint was studied in [26], while a frequency design objective function was constructed for multifrequency SAR with MI between SAR echo signal and target backscattering characteristics in [27].

As the target detection performance using SAR images directly reflects the target information acquisition capability of the corresponding SAR system [28], target detection MI (TDMI) is introduced in this work and a joint design of the frequency and bandwidth of the multifrequency SAR based on MI maximization is proposed. TDMI, describing the relationship between target existence hypothesis and target detection result, is derived and used to construct the objective function of the joint design. Then, the optimization problem for frequency and bandwidth joint design under engineering constraints is established and solved by the proposed modified genetic algorithm (GA) with code adjustment. Finally, computer simulation results are provided to verify the effectiveness of the proposed joint design method.

The remainder of this article is organized as follows. Section II builds the information acquisition model for multifrequency SAR and derives the TDMI expression. The constrained optimization problem is established and solved in Section III. The simulation results are provided in Section IV, and conclusions are drawn in Section V.

II. TDMI of Multifrequency SAR

In this section, the signal model of multifrequency SAR for target detection is presented first, and then the mathematical expression of TDMI is derived.

A. Signal Model for Multifrequency SAR Target Detection

Multifrequency SAR acquires target information by transmitting and receiving signals with different carrier frequencies and bandwidths. The received echo signal includes not only the target echo but also the background echo and the receiver noise. In analogy with the communication system [24], the target detection process of multifrequency SAR can be described by the model in Fig. 1. The target existence hypothesis and the target backscattering characteristics are considered as the source, the multifrequency SAR imaging process is the transmission channel, and the multifrequency images and the detection result are the sink. In this model, the target backscattering characteristics are reflected in SAR images, and the target existence hypothesis corresponds to the detection result.

The SAR system can be considered as working in a "go and stop" model. The echo is stored in a 2-D space with range and azimuth axes. The transfer function $\mathbf{x}(\tau, \eta)$ of multifrequency SAR echo acquisition can be regarded as a 2-D function of fast range time τ and slow azimuth time η, and it is expressed as

$$\mathbf{x}(\tau,\eta)=\mathbf{w_a}(\eta)\mathbf{w_r}\left(\tau-\frac{2R(\eta)}{c}\right)\exp\left\{-j4\pi\mathbf{f}\times\frac{R(\eta)}{c}\right\}\cdot\exp\left\{-j\boldsymbol{\varphi}\left(\tau-\frac{2R(\eta)}{c}\right)\right\} \quad (1)$$

where $\mathbf{f}=[f_1, f_2, \ldots, f_N]$ is the carrier frequency set, and $\mathbf{w_a}(\eta)$ and $\mathbf{w_r}(\tau)$ are the window functions in range and azimuth, respectively. The bandwidth set corresponding to the frequency set $\mathbf{f}$ is denoted by $\mathbf{b}=[b_1, b_2, \ldots, b_N]$. $R(\eta)$ is the range between the point target and the SAR platform. $\boldsymbol{\varphi}(\tau, \eta)$ is the phase of transmit signal at multiple frequencies. c is the speed of light. The boldfaced variables indicate vectors at different frequencies. The observed scene can be regarded as a superposition of target and background, and the backscattering of both is modulated on the transmit signals at N frequencies. The multifrequency SAR echo $\mathbf{s_{echo}}(\tau, \eta)$ received by the multifrequency receiver and demodulated to the baseband can be denoted as

$$\mathbf{s_{ehco}}(\tau,\eta)=\left\{\left[\mathbf{g}(\tau,\eta)+\mathbf{b}(\tau,\eta)\right]*\mathbf{x}(\tau,\eta)+\mathbf{n}(\tau,\eta)\right\}*\mathbf{r}(\tau,\eta) \quad (2)$$

where the stochastic processes $\mathbf{g}(\tau, \eta)$ and $\mathbf{b}(\tau, \eta)$ denote the target and the background backscattering characteristics at multiple frequencies, respectively, $\mathbf{n}(\tau, \eta)$ denotes the white Gaussian noise, and $\mathbf{r}(\tau, \eta)$ denotes the multifrequency receiver. The notation $*$ represents the 2-D convolution operator.

SAR imaging can be considered as an echo deconvolution process by a multifrequency imaging processor $\mathbf{h_{image}}(\tau, \eta)$. To suppress noise, the imaging processor is a correlation receiver of transmit signal. The convolution of the transmit signal, receiver, and imaging processor is a 2-D sinc function $\mathbf{h}(\tau, \eta)$ [29]. Thus, the multifrequency SAR image $\mathbf{s_{image}}(\tau, \eta)$ can be represented as

$$\begin{aligned}\mathbf{s_{image}}(\tau,\eta)&=\mathbf{s_{echo}}(\tau,\eta)*\mathbf{h_{image}}(\tau,\eta)\\&=\left[\mathbf{g}(\tau,\eta)+\mathbf{b}(\tau,\eta)\right]*\mathbf{h}(\tau,\eta)+\mathbf{n}(\tau,\eta)*\mathbf{u}(\tau,\eta)\end{aligned} \quad (3)$$

where $\mathbf{u}(\tau, \eta)=\mathbf{r}(\tau, \eta)*\mathbf{h_{image}}(\tau, \eta)$ denotes the convolution of the receiver and imaging processor.

Target detection in SAR images is usually performed in a sliding window. To circumvent the approximation issue caused by analog-to-digital conversion during signal processing [30], the discrete multifrequency SAR images patch $\mathbf{S_{image}}$ in a detection window of size $L_a \times L_r$ can be represented as

$$\mathbf{S_{image}}=\mathbf{H_a}[\mathbf{H_r}(\mathbf{G}+\mathbf{B})]^T+\mathbf{U_a}(\mathbf{U_r}\mathbf{N})^T \quad (4)$$

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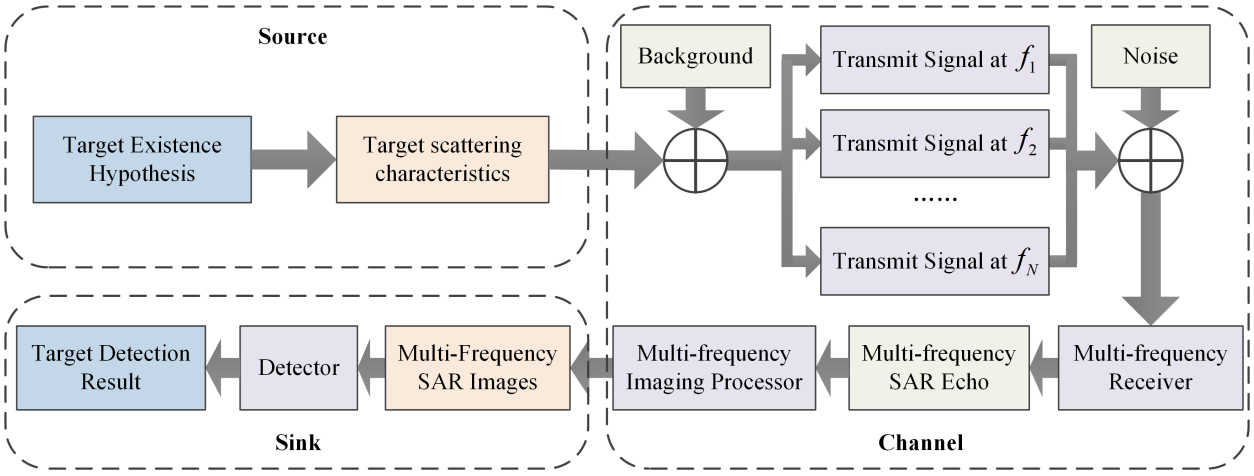


Fig. 1. Model for multifrequency SAR target detection.

where $\mathbf{S_{image}} \in \mathbb{C}^{N\times L_a\times L_r}$, $\mathbf{G} \in \mathbb{C}^{N\times K_r\times K_a}$, $\mathbf{B} \in \mathbb{C}^{N\times K_r\times K_a}$, and $\mathbf{N} \in \mathbb{C}^{N\times X_r\times X_a}$ are the discrete matrix forms of $\mathbf{s_{image}}(\tau,\eta)$, $\mathbf{g}(\tau,\eta)$, $\mathbf{b}(\tau,\eta)$, and $\mathbf{n}(\tau,\eta)$ in the detection window, respectively. $\mathbf{H_r} \in \mathbb{C}^{N\times L_r\times K_r}$, $\mathbf{H_a} \in \mathbb{C}^{N\times L_a\times K_a}$, $\mathbf{U_r} \in \mathbb{C}^{N\times L_r\times X_r}$, and $\mathbf{U_a} \in \mathbb{C}^{N\times L_a\times X_a}$ denote the Toeplitz matrices at multiple frequencies of $\mathbf{h_r} = [\mathbf{h_{r1}}, \mathbf{h_{r2}}, \ldots, \mathbf{h_{rM_r}}]^T$, $\mathbf{h_a} = [\mathbf{h_{a1}}, \mathbf{h_{a2}}, \ldots, \mathbf{h_{aM_a}}]^T$, $\mathbf{u_r} = [\mathbf{u_{r1}}, \mathbf{u_{r2}}, \ldots, \mathbf{u_{rZ_r}}]^T$, and $\mathbf{u_a} = [\mathbf{u_{a1}}, \mathbf{u_{a2}}, \ldots, \mathbf{u_{aZ_a}}]^T$ which are the discrete forms of sinc functions $\mathbf{h}(\tau)$ and $\mathbf{h}(\eta)$ and imaging processors $\mathbf{u}(\tau)$ and $\mathbf{u}(\eta)$ in range and azimuth dimensions, respectively. The relationships between the dimensions of these matrices are $L_r = K_r + M_r - 1 = X_r + Z_r - 1$ and $L_a = K_a + M_a - 1 = X_a + Z_a - 1$.

Using multifrequency SAR images, the target detection result D with detector $d\{\bullet\}$ in the detection window can be represented as

$$D = \begin{cases} d\{\mathbf{H_a}[\mathbf{H_r}(\mathbf{G}+\mathbf{B})]^T + \mathbf{U_a}(\mathbf{U_r}\mathbf{N})^T\}, & V = v_1 \\ d\{\mathbf{H_a}(\mathbf{H_r}\mathbf{B})^T + \mathbf{U_a}(\mathbf{U_r}\mathbf{N})^T\}, & V = v_0 \end{cases} \quad (5)$$

where the target existence hypothesis V has two possible values v_1 and v_0. v_1 indicates that the target exists in which case the pixels are a combined result of target, background, and noise; v_0 indicates that the target does not exist, in which case the pixels are the result of background and noise. D also has two possible values, d_1 indicating that the target is detected and d_0 indicating that the target is not detected.

B. MI of Multifrequency SAR Target Detection

With the target detection model in Fig. 1, the multifrequency SAR target detection can be considered as a binary channel since the source V and the sink D have two possible values. The average MI $I[D; V]$ between the hypothesis V and the detection result D can be expressed as (6), where $P_V(v_j)$ denotes the probability that the target existence hypothesis is v_j, and $P_{D|V}(d_i|v_j)$ denotes the conditional probability that the target detection result is d_i when $V = v_j$, and $i, j = 0$ or 1.

Therefore, when the prior probability $P_V(\bullet)$ of the target existence is known, the key to calculate $I[D; V]$ is the conditional probability $P_{D|V}(d_i|v_j)$ which is related to the detector $d\{\bullet\}$. Taking the hypothesis testing under the fusion criterion of minimizing Bayesian risk as an example, $P_{D|V}(d_i|v_j)$ is derived by carrying out the decision-level fusion on the detection results of SAR images at multiple frequencies

$$I[D; V] = -\sum_{i=0}^{1}\sum_{j=0}^{1} P_V(v_j)P_{D|V}(d_i|v_j) \cdot \log \frac{P_{D|V}(d_i|v_j)}{\sum_{j=0}^{1} P_V(v_j)P_{D|V}(d_i|v_j)}. \quad (6)$$

1) Multifrequency SAR Target Detection Based on Decision-Level Fusion: Let target detection result of N-frequency SAR images be $\mathbf{d} = [d^1, d^2, \ldots, d^n, \ldots, d^N]$. Under the assumption that the target scattering characteristics is independent at each frequency, the joint multifrequency target detection result D based on the minimized Bayesian risk fusion criterion for $\mathbf{d}$ can be expressed as

$$D = \begin{cases} d_0, & \prod_{n=1}^{N} P_{D|V,f_n,b_n}(d^n|v_0, f_n, b_n) \\ & \quad > \prod_{n=1}^{N} P_{D|V,f_n,b_n}(d^n|v_1, f_n, b_n) \\ d_1, & \prod_{n=1}^{N} P_{D|V,f_n,b_n}(d^n|v_0, f_n, b_n) \\ & \quad \le \prod_{n=1}^{N} P_{D|V,f_n,b_n}(d^n|v_1, f_n, b_n) \end{cases} \quad (7)$$

where $P_{D|V,f_n,b_n}(d^n|v_j, f_n, b_n)$ denotes the conditional probability that for the SAR image at the frequency f_n with bandwidth b_n, the target detection result is d^n when the target existence hypothesis is v_j, and $n \in [1, N]$.

For N-frequency SAR images, there are 2^N decision combinations. So, as shown in Fig. 2, each fusion decision result may correspond to multiple decision combinations. When the number of decision combinations leading to the fusion decision result $D = d_i$ is M_i and $i = 0$ or 1, we have $M_1 + M_0 = 2^N$. The mth combination can be expressed as the following equation:

$$\mathbf{d}^{i,m} = \left[\left(d^{i,m,1}|f_1, b_1\right), \ldots, \left(d^{i,m,n}|f_n, b_n\right), \ldots, \left(d^{i,m,N}|f_N, b_N\right)\right], \quad 1 \le m \le M_i \quad (8)$$

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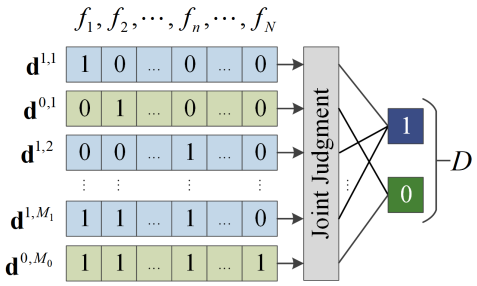


Fig. 2. Joint detection process.

where $d^{i,m,n}|f_n, b_n$ denotes the target detection result of SAR image at frequency f_n in the mth decision combination that leads to the fusion decision result $D = d_i$.

The conditional probability $P_{D|V}(d_i|v_j)$ of the fusion decision result is the sum of conditional probabilities $P_{\mathbf{d}|V}(\mathbf{d}^{i,m}|v_j)$ of M_i decision combinations which can be expressed as the product of the conditional probabilities $P_{D|V,f_n,b_n}(d^{i,m,n}|v_j, f_n, b_n)$ at multiple frequencies. Therefore, $P_{D|V}(d_i|v_j)$ can be expressed by $P_{D|V,f_n,b_n}(d^{i,m,n}|v_j, f_n, b_n)$ as follows:

$$\begin{aligned} P_{D|V}\left(d_i|v_j\right) &= \sum_{m=1}^{M_i} P_{\mathbf{d}|V}\left(\mathbf{d}^{i,m}|v_j\right) \\ &= \sum_{m=1}^{M_i}\prod_{n=1}^{N} P_{D|V,f_n,b_n}\left(d^{i,m,n}|v_j, f_n, b_n\right). \end{aligned} \tag{9}$$

Substituting (9) into (6), it can be seen that the TDMI of multifrequency SAR is related to the conditional probability of detection results at each frequency.

2) Conditional Probability of Single-Frequency Target Detection: Target detection in a single-frequency SAR image is achieved based on the detection statistics constructed by the image patch in the detection window. For an image patch $S_{\text{image}} \in \mathbb{C}^{L_a \times L_r}$ at frequency f_n, the pixels' sequence $\mathbf{s_w}$ is obtained by stretching the 2-D image patch S_{image} into a 1-D vector with length L_w. According to (4), $\mathbf{s_w}$ can be expressed as follows:

$$\begin{aligned} \mathbf{s_w} &= \text{vec}\left\{S_{\text{image}}\right\} \\ &= \text{vec}\left\{\text{H}_\text{a}[\text{H}_\text{r}(\text{G}+\text{B})]^T + \text{U}_\text{a}(\text{U}_\text{r}\text{N})^T\right\} \\ &= (\text{H}_\text{r}\otimes\text{H}_\text{a})\text{vec}\left\{\text{G}^T\right\} + (\text{H}_\text{r}\otimes\text{H}_\text{a})\text{vec}\left\{\text{B}^T\right\} \\ &\quad + (\text{U}_\text{r}\otimes\text{U}_\text{a})\text{vec}\left\{\text{N}^T\right\} \end{aligned} \tag{10}$$

where $\otimes$ denotes the Kronecker product, and vec($\bullet$) is the vectorization operator. Let $\text{H}_\text{T} \in \mathbb{C}^{L_w \times K_w} = \text{H}_\text{r} \otimes \text{H}_\text{a}$, $\text{U}_\text{T} \in \mathbb{C}^{L_w \times X_w} = \text{U}_\text{r} \otimes \text{U}_\text{a}$, $\mathbf{g_w} \in \mathbb{C}^{K_w \times 1} = \text{vec}\{\text{G}^T\}$, $\mathbf{b_w} \in \mathbb{C}^{K_w \times 1} = \text{vec}\{\text{B}^T\}$, and $\mathbf{n_w} \in \mathbb{C}^{X_w \times 1} = \text{vec}\{\text{N}^T\}$, where $L_w = L_a L_r$, $K_w = K_a K_r$, and $X_w = X_a X_r$. Thus, (10) can be simplified as

$$\mathbf{s_w} = \text{H}_\text{T}\mathbf{g_w} + \text{H}_\text{T}\mathbf{b_w} + \text{U}_\text{T}\mathbf{n_w}. \tag{11}$$

To show the effect of b_n on target detection, the target backscattering sequence $\mathbf{g_w}$ is regarded as the concatenation of a real target sequence of length K'_w and a zero sequence of length $K_w - K'_w$. K'_w is the total number of sampling points in the target area in the detection window and can be represented as $K'_w = K'_r \times K'_a$, where K'_r and K'_a denote the number of sampling points in the target area in range and azimuth, respectively. K'_a is constant under a fixed azimuth resolution. K'_r is determined by the target size l_{target} in the range dimension and bandwidth b_n, and it can be expressed as

$$K'_r = \frac{2b_n l_{\text{target}}}{c}. \tag{12}$$

Therefore, $\mathbf{g_w}$ and $\mathbf{b_w}$ can be represented as

$$\begin{cases} \mathbf{g_w} = \left[g_1, g_2, \ldots, g_{K'_w}, 0, \ldots, 0\right]^T \\ \mathbf{b_w} = \left[b_1, b_2, \ldots, b_{K'_w}, b_{K'_w+1}, \ldots, b_{K_w}\right]^T. \end{cases} \tag{13}$$

The target detection model in (5) can be represented as

$$\begin{cases} v_1 : \mathbf{s_w} = \text{H}_\text{T}\mathbf{g_w} + \text{H}_\text{T}\mathbf{b_w} + \text{U}_\text{T}\mathbf{n_w} \\ v_0 : \mathbf{s_w} = \text{H}_\text{T}\mathbf{b_w} + \text{U}_\text{T}\mathbf{n_w}. \end{cases} \tag{14}$$

It is assumed that the target and the background area are homogeneous and obey the complex Gaussian distribution with zero mean [31]. The noise is assumed to be complex white Gaussian. The target, background, and noise are independent of each other. The covariance matrices of $\mathbf{s_w}$ at frequency f_n under the hypotheses v_1 and v_0 can be represented as

$$\begin{aligned} R_{\mathbf{s}0|f_n,b_n} &= \text{H}_\text{T}E\left[\mathbf{b_w}\mathbf{b_w}^H\right]\text{H}_\text{T}{}^H + \text{U}_\text{T}E\left[\mathbf{n_w}\mathbf{n_w}^H\right]\text{U}_\text{T}{}^H \\ &= \text{H}_\text{T}R_{\mathbf{b}|f_n,b_n}\text{H}_\text{T}{}^H + \text{U}_\text{T}R_{\mathbf{n}|f_n,b_n}\text{U}_\text{T}{}^H \end{aligned} \tag{15}$$

$$\begin{aligned} R_{\mathbf{s}1|f_n,b_n} &= \text{H}_\text{T}E\left[\mathbf{g_w}\mathbf{g_w}^H\right]\text{H}_\text{T}{}^H + R_{\mathbf{s}0|f_n,b_n} \\ &= \text{H}_\text{T}R_{\mathbf{g}|f_n,b_n}\text{H}_\text{T}{}^H + R_{\mathbf{s}0|f_n,b_n} \end{aligned} \tag{16}$$

where $R_{\mathbf{g}|f_n,b_n}$, $R_{\mathbf{b}|f_n,b_n}$, and $R_{\mathbf{n}|f_n,b_n}$ denote, respectively, the covariance matrices of target, background, and noise at frequency f_n which can be calculated by the power spectrum.

The conditional probability of the image patch in detection window can be formulated as

$$\begin{cases} P(\mathbf{s_w}|v_1, f_n, b_n) \\ \quad = \dfrac{1}{\pi^L \det\left(R_{\mathbf{s}1|f_n,b_n}\right)} \exp\left[-\mathbf{s_w}^H R^{-1}_{\mathbf{s}1|f_n,b_n}\mathbf{s_w}\right] \\ P(\mathbf{s_w}|v_0, f_n, b_n) \\ \quad = \dfrac{1}{\pi^L \det\left(R_{\mathbf{s}0|f_n,b_n}\right)} \exp\left[-\mathbf{s_w}^H R^{-1}_{\mathbf{s}0|f_n,b_n}\mathbf{s_w}\right]. \end{cases} \tag{17}$$

The discriminant under the minimum error probability criterion is

$$L(\mathbf{s_w}) = \ln\frac{P(\mathbf{s_w}|v_1, f_n, b_n)}{P(\mathbf{s_w}|v_0, f_n, b_n)} \underset{v_0}{\overset{v_1}{\gtrless}} \ln\frac{P_V(v_0)}{P_V(v_1)} = \ln\left(\frac{1-\alpha}{\alpha}\right). \tag{18}$$

Therefore, the conditional probability of target detection at the frequency f_n can be expressed by

$$\begin{aligned} &P_{D|V,f_n,b_n}(d_i|v_j, f_n, b_n) \\ &\quad = \int_{\mathcal{Z}_i} P\left(\ln\frac{P(s_w|v_1, f_n, b_n)}{P(s_w|v_0, f_n, b_n)}\middle| v_j\right) ds_w \end{aligned} \tag{19}$$

where $\mathcal{Z}_i$ denotes the verdict of $\mathbf{s_w}$ leading to the decision d_i. Detailed derivation of (19) can be found in Appendix.

The conditional probability $P_{D|V}(d_i|v_j)$ of fusion detection can be calculated by substituting (19) into (9). By further

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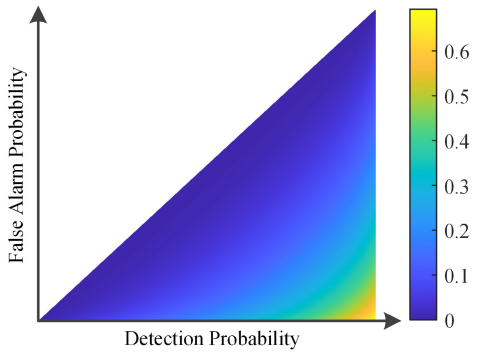


Fig. 3. Relationship between TDMI and detection performance.

substituting $P_{D|V}(d_i|v_j)$ into (6), the TDMI of multifrequency SAR can be expressed as

$$I[D;V] = -\sum_{i=0}^{1}\sum_{j=0}^{1}\mathcal{C}\left(d_i, v_j, \alpha, \mathbf{f}, \mathbf{b}\right) \tag{20}$$

where

$$\begin{aligned}&\mathcal{C}\left(d_i, v_j, \alpha, \mathbf{f}, \mathbf{b}\right)\\ &= P_V(v_j)\int_{\mathcal{Z}_i} P\left(\ln\frac{P(s_w|v_1, f_n, b_n)}{P(s_w|v_0, f_n, b_n)}\middle| v_j\right) ds_w\\ &\quad\cdot\log\left[\frac{\int_{\mathcal{Z}_i} P\left(\ln\frac{P(s_w|v_1, f_n, b_n)}{P(s_w|v_0, f_n, b_n)}\middle| v_j\right) ds_w}{\sum_{j=0}^{1} P_V(v_j)\int_{\mathcal{Z}_i} P\left(\ln\frac{P(s_w|v_1, f_n, b_n)}{P(s_w|v_0, f_n, b_n)}\middle| v_j\right) ds_w}\right].\end{aligned} \tag{21}$$

It is shown that the TDMI is related to the a priori probability of target existence and the frequencies and bandwidths of transmit signals.

It can be seen that $P_{D|V}(d_i|v_j)$ can be used to describe the target detection performance. When $D = d_1$ and $V = v_1$, $P_{D|V}(d_1|v_1)$ is the detection probability that the target exists and has been detected. When $D = d_1$ and $V = v_0$, $P_{D|V}(d_1|v_0)$ is the false alarm probability that the target does not exist but has been detected. Similarly, $P_{D|V}(d_0|v_1)$ is the miss probability and $P_{D|V}(d_0|v_0)$ is the correct rejection probability. So, the TDMI of multifrequency SAR is related to the target detection performance. The relationship between TDMI, detection probability, and false alarm probability is shown in Fig. 3 where the maximum of TDMI appears when the detection probability is 1 and the false alarm probability is 0, indicating the TDMI embodies both the detection probability and false alarm probability. Thus, TDMI can comprehensively represent the target detection performance of SAR and multifrequency SAR can be optimized when the TDMI is maximized.

III. Joint Design Method of Frequency and Bandwidth for Multifrequency SAR

In this section, a joint design method for frequency and bandwidth is presented based on MI maximization. The optimization problem is established with the objective function constructed by TDMI under some constraints of multifrequency SAR. A modified GA with code adjustment operator is proposed to solve it.

A. Formulation of the Constrained Optimization Problem

Based on MI maximization, the optimization objective function for $\mathbf{f}$ and $\mathbf{b}$ can be constructed by TDMI according to (20) as follows:

$$\begin{aligned}\hat{\mathbf{f}}, \hat{\mathbf{b}} &= \arg\max_{\mathbf{f},\mathbf{b}} I[D;V]\\ &= \arg\max_{\mathbf{f},\mathbf{b}}\left\{-\sum_{i=0}^{1}\sum_{j=0}^{1}\mathcal{C}\left(d_i, v_j, \alpha, \mathbf{f}, \mathbf{b}\right)\right\}.\end{aligned} \tag{22}$$

To make the designed frequencies and bandwidths realizable in practice, it is necessary to impose some reasonable constraints on the design, as detailed in the following.

Transmit Power: Due to the limitation of total transmit signal power, the number of frequencies and the total bandwidth are limited, i.e., $1 \le N \le N_{\max}$ and $\sum_{n=1}^{N} b_n \le B_{\text{sum}}$.

Frequency Range: To ensure high-quality transmission of SAR signals, the carrier frequencies should avoid peak atmospheric absorption regions [32] and existing RF interference. Defining the feasible frequency range excluding the above ranges as $\mathcal{R}_f$, this constraint can be formulated as $f_n \in \mathcal{R}_f$.

Range Resolution: To acquire SAR images with high range resolution, the lower bound on the bandwidth of the transmit signal at each frequency is required to be $B_{\min}$. The maximum bandwidth limited by hardware is set to $B_{\max}$. This constraint can be formulated as $B_{\min} \le b_n \le B_{\max}$.

Redundancy of Target Characteristics: It is necessary to minimize the redundancy of target characteristics between the selected bands. The constraint involves the following two aspects. First, the selected bands should not overlap. Let the band with the frequency f_n be $B_n = [f_n - b_n/2, f_n + b_n/2]$. For $\forall n, k \in [1, N]$ and $n \ne k$, this constraint is given as $B_n \cap B_k = \emptyset$. Second, the target backscattering characteristics at selected frequencies should be considered as noncorrelated. The correlation of the target backscattering characteristics at each frequency can be described by the frequency correlation function (FCF) [33], which is solely dependent on the frequency interval. The redundancy between neighboring frequencies can be considered small enough if the interval is greater than Δf_{FCF}. So, the constraint can be formulated as $f_{n+1} - f_n > \Delta f_{\text{FCF}}$, where $1 \le n \le N-1$.

Considering all the above constraints, the constrained optimization problem for joint frequency and bandwidth design based on MI maximization is formulated as

$$\begin{aligned}\hat{\mathbf{f}}, \hat{\mathbf{b}} = \arg\max_{\mathbf{f},\mathbf{b}}&\left\{-\sum_{i=0}^{1}\sum_{j=0}^{1}\mathcal{C}\left(d_i, v_j, \alpha, \mathbf{f}, \mathbf{b}\right)\right\}\\ \times&\begin{cases}1 \le N \le N_{\max}\\ \sum_{n=1}^{N} b_n \le B_{\text{sum}}\\ f_n \in \mathcal{R}_f, \quad 1 \le n \le N\\ B_{\min} \le b_n \le B_{\max}, \quad 1 \le n \le N\\ B_n \cap B_k = \emptyset, \quad 1 \le n, k \le N, \ n \ne k\\ f_{n+1} - f_n > \Delta f_{\text{FCF}}, \quad 1 \le n \le N-1.\end{cases}\end{aligned} \tag{23}$$

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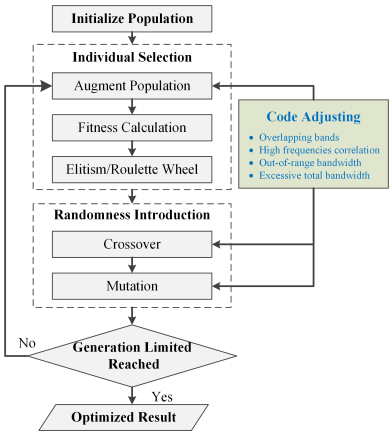


Fig. 4. Flowchart of the modified GA.

B. Solution Using Modified Genetic Algorithm

It is difficult to obtain a global optimal solution by traditional mathematical programming methods for the problem in (23) because of the unanalyzable continuity and differentiability of (22) [34]. Therefore, the intelligent optimization method is considered to solve (23).

The joint design of frequency and bandwidth can be described as a combinatorial optimization problem that sparsely allocates bandwidth resources over a broad frequency range. GA, as a heuristic intelligent optimization method, simulates the genetic selection and inheritance of organisms under the rule of survival of the fittest. It is chosen to solve (23) as it is suitable for searching for the optimal solution in large-scale combinatorial spaces.

The most widely used approach to deal with constrained optimization problems in GA is the penalty function method, which adds penalty factors and penalty function to the objective function [35]. However, the optimal solution of the objective function with the penalty function may not be that of the original problem when solving the constrained optimization problem with multiple constraints like (23).

Therefore, a modified GA is proposed in this section by adding a code adjustment operator to ensure that the individuals in evolution are always in the feasible region satisfying the constraints. As shown in Fig. 4, the initial population created randomly generates the next-generation population through the selection operation and the introduction of randomness, and any individuals outside the feasible region generated in this process need to be adjusted by the code adjustment operator. Repeat the above process until the maximum number of iterations is reached. The optimized frequencies and bandwidth are obtained by decoding the individual with the best fitness in the last generation population.

The detailed steps are as follows.

Step 1 (Population Initialization): Individual code consists of a selected frequencies sequence and a corresponding bandwidths' sequence, as given in (24). The maximum number of frequencies $N_{\max}$ in the constraint of transmit power determines that the length of the code is $2N_{\max}$. When the number of selected frequencies N_{select} is less than $N_{\max}$, the $N_{\text{select}}+1$th to $N_{\max}$th elements of the frequencies and bandwidths sequence are set to zero, respectively,

$$[f_1, f_2, \ldots, f_{N_{\max}}, b_1, b_2, \ldots, b_{N_{\max}}]. \tag{24}$$

For each individual, a dynamic library of available frequencies and bandwidths is constructed based on constraints, and the initial population is generated by randomly selecting frequency–bandwidth pairs $[f_n, b_n]$ from it. Each time when a $[f_n, b_n]$ is selected, the dynamic library changes according to the constraints until $N_{\max}$ frequency–bandwidth pairs have been selected, or the available bandwidths do not satisfy the minimum resolution constraint.

Step 2 (Individual Selection): To avoid the GA being trapped into a local optimum, an augmentation and contraction method proposed by [36] is introduced. For the code given by (24), the augmentation operation circularly shifts the nonzero parts of the frequency and bandwidth sequences, respectively, and fuses the original sequences and the shifted sequences with randomized weights. The augmentation increases the size of the population from N_p to $2N_p$ and improves the global search ability of GA by expanding the selection space.

The fitness function of individuals in the augmented population with size $2N_p$ is calculated by TDMI according to (20). Then, a selection strategy that combines elitist and roulette wheel is used to select better individuals in a population, which preserves the individuals with optimal fitness in the augmented population and selects the others by the roulette wheel.

Step 3 (Randomness Introduction): In this step, crossover and mutation are performed on individuals with a certain probability. The crossover exchanges two individual codes from a random position, and the mutation changes the individual code at a random position. Since the codes contain one-to-one frequencies and bandwidths, these operations are performed on the frequency sequences in code, and the bandwidth sequence changes with frequency sequence.

Step 4 (Code Adjustment): Due to the randomness introduced by the population augmentation, crossover, and mutation, individuals may be created outside the feasible region and need to be adjusted to satisfy the multiple constraints. The adjustments should follow the principle that later adjustments do not affect constraints that have already been satisfied. The detailed steps are as follows.

1) Adjustments for overlapping bands. This adjustment fuses two overlapping bands and sets the new selected frequency to the center of the fused band. The fusion bandwidth may exceed the maximum value $B_{\max}$, which is adjusted in 3).
2) Adjustment for high frequencies' correlation. If the interval of two neighboring frequencies is less than Δf_{FCF}, the larger one is randomly shifted to any optional frequency that satisfies the constraint.
3) Adjustment for out-of-range bandwidth. Bandwidths larger than the maximum value are replaced by $B_{\max}$. To avoid the total adjusted bandwidth exceeding $B_{\max}$, bandwidths smaller than the minimum value $B_{\min}$ are discarded.
4) Adjustment for excessive total bandwidth. When the sum of the bandwidths exceeds B_{sum}, the excess bandwidth b_e is proportionally subtracted from the current bandwidth. For bandwidth b_i, the ratio η_i is calculated by (25), and

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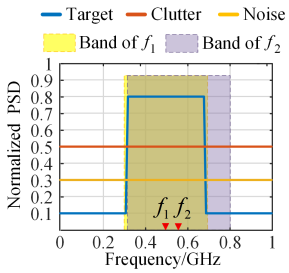


Fig. 5. Power spectrum of target, background, and noise.

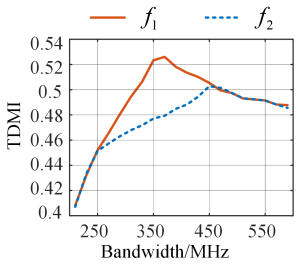


Fig. 6. TDMI with bandwidth at two carrier frequencies.

the adjusted bandwidth is expressed as $b_i' = b_i - \eta_i b_e$, where K_b is the number of nonzero bandwidths

$$\eta_i = \frac{b_i - B_{\min}}{\sum_{i=1}^{K_b} (b_i - B_{\min})}. \tag{25}$$

The code adjustment operation not only makes the individuals satisfy the multiple constraints but also introduces randomness into the population by the random relocation of frequency in 2), helping GA find a globally optimal solution in the feasible region.

IV. Simulation Results

In this section, simulations are conducted to demonstrate the effectiveness of the proposed design. First, the influence of frequency and bandwidth on TDMI is illustrated. Second, the multiple frequencies and bandwidth are designed for the target with a specific nonuniform power spectrum to show the feasibility of the proposed method. Finally, the designed frequencies and bandwidths for a real scene are used for imaging simulation and target detection. By comparing with the existing spaceborne SAR, the effectiveness of the proposed method in improving the target information acquisition capability and the target detection performance of multifrequency SAR is verified. To simplify the calculation, the azimuth length of the detection window is set to 1 in the simulations, and the range length is determined by the target size and the maximum bandwidth.

A. Influence of Frequency and Bandwidth on TDMI

The variation in TDMI with bandwidth for a specific target at two different frequencies is presented. The target normalized power spectrum density (PSD) is shown by the blue line in Fig. 5, whose amplitude is 0.8 from 325 to 675 MHz and 0.1 for the other frequencies. Assuming that the background and the noise are both white Gaussian with a power of 0.5 and 0.3, their normalized power spectrums are shown by the red and yellow curves in Fig. 5, respectively.

When the carrier frequency is $f_1 = 500$ MHz, the TDMI under the bandwidth from 200 to 600 MHz is calculated by (20) and shown by the red solid line in Fig. 6. The change in TDMI with bandwidth can be divided into three stages: first, TDMI increases rapidly when the bandwidth is less than 350 MHz; second, it increases slowly with the bandwidth from 350 to 370 MHz and reaches the maximum value at 370 MHz, and then it decreases when the bandwidth is larger than 370 MHz. The rapid increase in TDMI in the first stage is because the power spectrum amplitude of the target is always larger than that of the background and noise in the band of the transmit signal and the received target energy increases more than the background and noise. When the bandwidth exceeds 350 MHz, part of the target power spectrum is submerged with background and noise in the band and the received energy increment of target is less than that of the background and noise. However, a larger bandwidth leads to higher resolution, which makes the target occupy more pixels and easier to be detected. Thus, in the second stage the TDMI slowly increases as the influence of high resolution is greater than that of the submerged target power spectrum. In the third stage with the bandwidth exceeding 370 MHz, compared with high resolution, the influence of submerged target power dominates, making the TDMI decrease. The analysis above shows that the bandwidth which gives the maximum TDMI is determined by both the resolution and the power spectrum of target, background, and noise, and the optimal bandwidth of f_1 is shown by the yellow part in Fig. 5.

When the carrier frequency is $f_2 = 550$ MHz, TDMI changes with the bandwidth are shown by the blue dashed line in Fig. 6. The change can also be divided into three stages: first, it increases rapidly when the bandwidth is less than 250 MHz, then increases slowly with the bandwidth from 250 to 450 MHz and reaches a maximum value at 450 MHz, and finally decreases when the bandwidth exceeds 450 MHz. The optimal bandwidth of f_2 is shown by the purple part in Fig. 5. The changes in the first and third stages of f_2 are due to the same reason as in the f_1 case. In the second stage with the bandwidth from 250 to 450 MHz, the power spectrum amplitude of target is higher than that of the background and noise at the left part of the band but is lower in the right part, leading to a slow increase in TDMI. The band with a width of 450 MHz which makes the TDMI maximum can be divided into two parts: one with a width of 350 MHz corresponding to the target spectrum above the background and noise, and the other with a width of 100 MHz corresponding to the target spectrum under the background and the noise.

Comparing the TDMI of f_1 and f_2, it can be seen that TDMI of f_1 is not only larger than that of f_2 at their peaks but also larger than or equal to that of f_2 under any bandwidth. As a result, for the scene with the power spectrum in Fig. 5, a transmit signal with a bandwidth of 370 MHz and carrier frequency of f_1 can obtain the maximum TDMI. In summary, for a specific target, the design of frequency and bandwidth plays a significant role in improving the TDMI of SAR.

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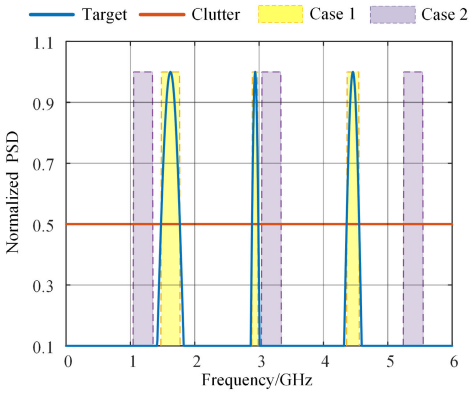


Fig. 7. Normalized power spectrum of scene and multifrequency emission cases.

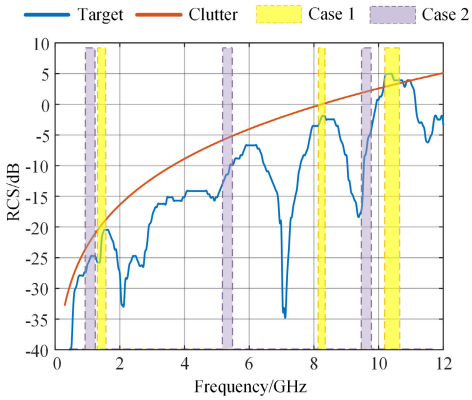


Fig. 8. Normalized power spectrum of scene and multifrequency emission cases.

TABLE I
PARAMETERS IN THE JOINT DESIGN

Parameter	Value
$N_{\max}$	3
B_{sum}	900MHz
$\mathcal{R}_f$	0-6GHz
$\mathcal{R}_b$	50-500MHz
Δf_{FCF}	0.7GHz
P_{cross}	0.5
P_{mutate}	0.5
N_p	100
P_{target}	0.5
l_{target}	50m

TABLE II
DETECTION PERFORMANCE AND TDMI OF TWO CASES

Case	f/GHz	b/MHz	TDMI	Joint P_d	Joint P_f
The Proposed Method	1.62	290	0.33	0.86	0.09
	2.94	90			
	4.46	190			
Experienced	1.20	300	0.04	0.61	0.33
	3.20	300			
	5.40	300			

B. Joint Design of Multiple Frequencies and Bandwidths for a Specific Scene

To verify the effectiveness of the proposed method, three frequencies and bandwidths are jointly designed for a specific scene. The normalized power spectrum of the target with the power concentrated around $f_1 = 1.62$ GHz, $f_2 = 2.94$ GHz, and $f_3 = 4.46$ GHz is shown by the blue line in Fig. 7, and the normalized power spectrum of background is shown by the red line. The transmit signal is LFM and the receiver noise is white Gaussian with the same power spectrum as the background. The main parameters used in the joint design are given Table I. The crossover probability P_{cross} and mutation probability P_{mutate} in GA are both 0.5, and the population size N_p is 100. The prior probability of target existence P_{target} is 0.5, and the target size l_{target} is 50 m.

The designed frequencies and bandwidths obtained by 500 iterations of GA are given in Table II, and the transmit signal band is shown by the yellow part named Case 1 in Fig. 7. As a comparison, the bands of the signal with the existing empirical carrier frequencies and bandwidths are given by the purple part named Case 2 in Fig. 7. These empirical frequencies are 1.2 GHz in the L-band, 3.2 GHz in the S-band, and 5.4 GHz in the C-band, and the bandwidth of each frequency is 300 MHz. Clearly, compared with Case 2, Case 1 with the designed frequencies and bandwidths is more compatible with the target power spectrum and the transmit energy concentrates at the bands with higher amplitude of the target power spectrum.

To show the advantages of the designed frequencies and bandwidths in target detection performance, the TDMI, target detection probability, and false alarm probability of multi-frequency SAR with the parameters in Cases 1 and 2 are calculated by (20) and (9). The results, listed in Table II, show that both the TDMI and joint detection performance are improved by the proposed method compared with the empirical Case 2. In addition, the total designed bandwidth 570 MHz is much smaller than 900 MHz of Case 2. Therefore, the proposed method not only achieves higher target information acquisition capability but also saves transmit resources.

C. Joint Design of Frequency and Bandwidth for an Airplane Target

To further demonstrate the effectiveness of the proposed method, the frequencies and bandwidths are designed for an airplane target, and imaging simulation and target detection are performed with the designed results.

Take a canard airplane as the target and ground clutter as the background. The RCS of the target and the background is shown in Fig. 8, where the blue line presents the RCS of the aircraft changing with frequency under a look angle of 45° and HH polarization [37], and the red line is the RCS of the ground clutter computed by the model $\gamma - f$ [38]. The normalized power spectrum $H_1(j\omega)$ of the target during wideband observation can be calculated from the RCS σ, i.e., $|H_1(j\omega)|^2 = (\sigma/\pi c^2)$ [39]. The transmit signal is LFM, and the constraints and parameters in joint design are similar to those in Table I, except that the frequency range is changed to $\mathcal{R}'_f = [300\ \text{MHz}, 12\ \text{GHz}]$. The designed results are obtained by 500 iterations of GA and listed in Table III. The designed frequencies are 1.55, 8.28, and 10.43 GHz, and the corresponding bandwidths are 250, 200, and 450 MHz. The designed band of the transmit signal is shown by yellow part named Case 1 in Fig. 8.

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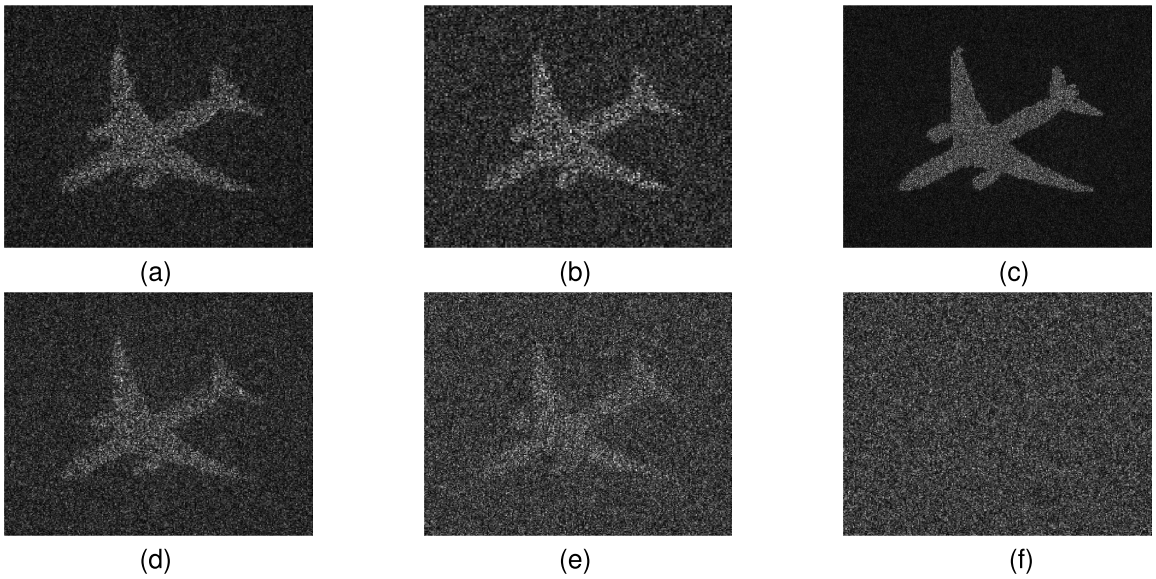


Fig. 9. Simulation images with designed results and existing SAR parameters. (a) 1.55 GHz. (b) 8.28 GHz. (c) 10.43 GHz. (d) 1.20 GHz. (e) 5.40 GHz. (f) 9.65 GHz.

TABLE III
PERFORMANCE INDICATORS OF DESIGNED RESULTS AND EXISTING SAR SYSTEMS

		f/GHz	b/MHz	SCR	TDMI
Results Designed by Proposed Method		1.55	250	4.28	0.71
		8.28	200	3.18	
		10.43	450	10.09	
Existing SAR	LT-1(L-band)	1.20	300	2.85	0.62
	Hisea-1(C-band)	5.40	300	1.77	
	TerraSAR(X-band)	9.65	300	1.11	

Three existing spaceborne SAR systems at different bands are given as a comparison, which are LT-1 at the L-band [40], Hisea-1 at the C-band [41], and TerraSAR at the X-band [42]. Their frequencies and bandwidths are listed in Table III, and the band of the transmit signal is shown by purple part named Case 2 in Fig. 8. The TDMI of both the cases is calculated by (20) and listed in Table III. It is shown that compared with the TDMI of 0.6175 for the existing SAR systems, the multifrequency SAR with designed parameters can obtain a higher TDMI of 0.7058.

To intuitively demonstrate the advantages of designed frequencies and bandwidths in target detection performance, the SAR system parameters in Table III are used for imaging simulation and target detection. The scattering coefficients of the target and background in the scene are different samples of one random process. First, three data matrices are generated according to the statistics of target, background, and noise. Then, delimiting the area for airplane, the associated cells of target data matrix are kept, and the observed scene is formed with the associated ground clutter components. The ratio of the target normalized power spectrum mean value to the white Gaussian noise normalized power is set to 0 dB. Echo simulation [43] and RD imaging [44] are performed at the designed results and empirical parameters. It should be noted that the imaging process must take into account the difference in resolution due to the bandwidth. The simulation images are shown in Fig. 9.

The signal-to-clutter ratio (SCR) of the images at each frequency is shown in the penultimate column of Table III. It can be seen that the results in Fig. 9(a)–(c) obtained with designed parameters have higher SCR than those in Fig. 9(e)–(g) with the existing spaceborne SAR parameters.

Then target detection is performed on the simulation SAR images in Fig. 9 and the results are shown in Fig. 10. It can be seen that all three simulated images with designed frequencies and bandwidths correctly detect the target, while only one of the three images using the existing parameters has detected the target correctly at 1.2 GHz.

The imaging and detection simulation is repeated 100× under the hypotheses of target existence and nonexistence, respectively, with SNR from −15 to 30 dB, and the detection probability and the false alarm probability of a single-frequency image with each SNR are calculated by the ratio of the detected samples count to the total samples count of two hypotheses. The SNR is defined as the ratio of the target power spectrum mean value to the white noise power. The detection probability and the false alarm probability of joint detection are calculated by (9). The red solid line and the blue dashed line in Fig. 11 show the detection probability of the designed results and the existing parameters with SNR, respectively. As SNR increases, the detection probability of the designed results gradually increases and reaches 1 at the SNR of −3 dB. In contrast, the detection probability of the existing parameters reaches a maximum value of 0.9 at 25 dB and is lower than that of the designed results at any SNR.

The red solid line and the blue dashed line in Fig. 12 show the false alarm probability in two cases. As SNR increases, the false alarm probability of the designed results decreases gradually and drops to almost 0 at an SNR of 0 dB. In contrast, the false alarm probability with the existing parameters decreases to a stable value of 0.04 when the SNR exceeds 10 dB and is higher than that of the designed results at any SNR.

It can be seen that compared with the existing parameters, higher detection accuracy can be achieved at lower

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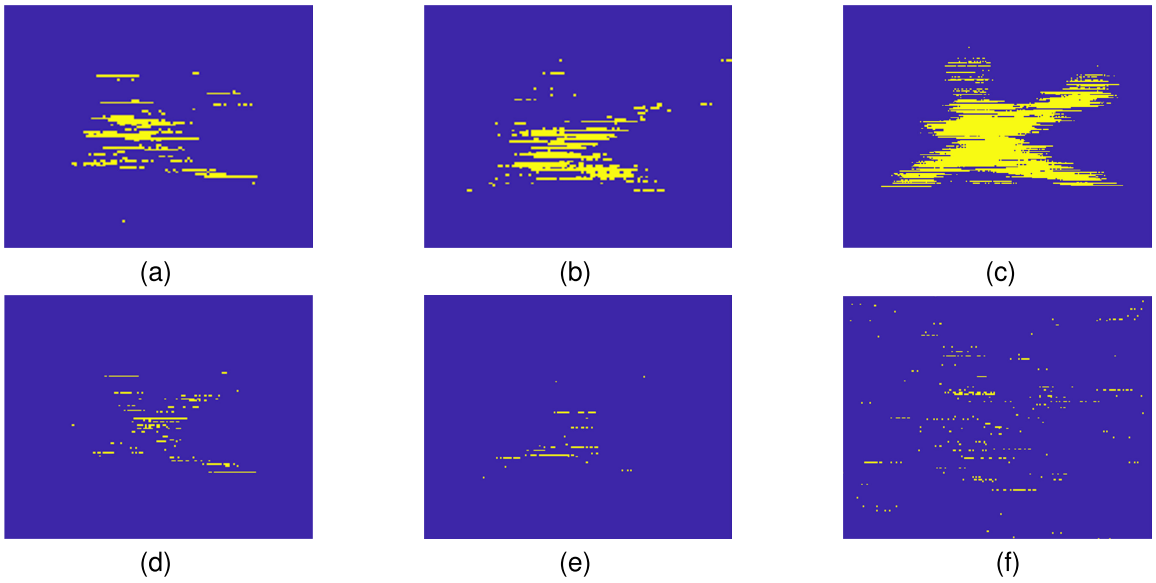


Fig. 10. Target detection result of simulation images. (a) 1.55 GHz. (b) 8.28 GHz. (c) 10.43 GHz. (d) 1.20 GHz. (e) 5.40 GHz. (f) 9.65 GHz.

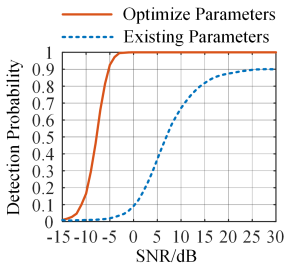


Fig. 11. Detection probability with SNR.

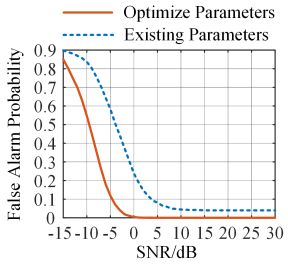


Fig. 12. False alarm probability with SNR.

SNR when target detection is performed with the designed parameters.

The image simulation and detection results confirm that the multifrequency SAR with frequency and bandwidth designed by the proposed method has higher target information acquisition capability than the existing SAR systems and can provide high-quality SAR images and accurate target detection results.

V. Conclusion

In this article, a joint frequency and bandwidth design method for multifrequency SAR based on MI maximization has been presented. The optimization problem for joint frequency and bandwidth design was first established under some engineering constraints by maximizing TDMI. Then, a modified GA with code adjustment was proposed to solve the problem and obtain the frequencies and bandwidths with high information acquisition and detection accuracy for a specific scene. Finally, performance of the designed results and the empirical parameters was evaluated in terms of information acquisition, imaging, and target detection. Simulation results have shown that the multifrequency SAR with designed frequencies and bandwidths is able to obtain greater TDMI, larger SCR of image, higher detection probability, and lower false alarm probability than the existing spaceborne SARs.

Appendix
Conditional Probability of Target Detection in SAR Image at Single Frequency

The hypothesis testing model and statistical property of SAR image patch in the detection window at frequency f_n are given in Section II-B2. According to (18), $L(\mathbf{s_w})$ can be written as

$$L(\mathbf{s_w}) = -\ln \frac{\det\left(\mathrm{H_T} R_{\mathbf{g}|f_n,b_n} \mathrm{H_T}^H + R_{\mathbf{s0}|f_n,b_n}\right)}{\det\left(R_{\mathbf{s0}|f_n,b_n}\right)} + \mathbf{s_w}^H \times \left[R^{-1}_{\mathbf{s0}|f_n,b_n} - \left(\mathrm{H_T} R_{g|\mathrm{f}_n,b_n} \mathrm{H_T}^H + R_{\mathbf{s0}|f_n,b_n}\right)^{-1}\right] \mathbf{s_w}. \tag{26}$$

The expression of (26) is too complex to derive the probability distribution of the test statistic and should be simplified. Let $A = R^{-1/2}_{\mathbf{s0}|f_n,b_n} \mathrm{H_T} R_{\mathbf{g}|f_n,b_n} \mathrm{H_T}^H R^{-1/2}_{\mathbf{s0}|f_n,b_n} + R_{\mathbf{s0}|f_n,b_n}$, and we perform an eigenvalue decomposition of A, i.e., $A = Q\Lambda Q^H$, where Q is the eigenvector matrix, and $\Lambda = \mathrm{diag}(\lambda_1, \lambda_2, \ldots, \lambda_{L_w})$ is a diagonal matrix composed of corresponding eigenvalues. Thus, (26) can be rewritten as

$$L(\mathbf{s_w}) = -\det(I + \Lambda) + \mathbf{s_w}^H R^{-1/2}_{\mathbf{s0}|f_n,b_n} \times Q^H \left[I - (I + \Lambda)^{-1}\right] Q R^{-1/2}_{\mathbf{s0}|f_n,b_n} \mathbf{s_w} \tag{27}$$

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where $I \in \mathbb{C}^{L_w \times L_w}$ is the unit matrix. Let $\mathbf{z} = Q R_{\mathbf{s}0|f_n,b_n}^{-1/2} \mathbf{s}_\mathbf{w}$, and (26) can be rewritten as a new one with $\mathbf{z}$ as

$$L(\mathbf{s}_\mathbf{w}) = L(\mathbf{z}) = -\det(I+\Lambda) + \mathbf{z}^H \left[I - (I+\Lambda)^{-1} \right] \mathbf{z}. \quad (28)$$

Substituting $\Lambda = \mathrm{diag}(\lambda_1, \lambda_2, \ldots, \lambda_{L_w})$ into (28), we obtain

$$L(\mathbf{z}) = -\sum_{i=1}^{L_w} \ln(1+\lambda_i) + \sum_{i=1}^{L_w} z_i^2 \frac{\lambda_i}{1+\lambda_i}. \quad (29)$$

Based on the minimum total error probability criterion, the discriminant can be recast as

$$\begin{aligned} L(\mathbf{s}_\mathbf{w}) &= \ln \frac{P(\mathbf{s}|v_1, f_n, b_n)}{P(\mathbf{s}|v_0, f_n, b_n)} \\ &= \ln \frac{P(\mathbf{z}|v_1, f_n, b_n)}{P(\mathbf{z}|v_0, f_n, b_n)} = L(\mathbf{z}) \underset{v_0}{\overset{v_1}{\gtrless}} \ln \frac{P_V(v_0)}{P_V(v_1)}. \end{aligned} \quad (30)$$

We have $P(\mathbf{z}|v_0) \sim \mathrm{CN}(0, I)$ and $P(\mathbf{z}|v_1) \sim \mathrm{CN}(0, I+\Lambda)$ by observing (17), which means $z_i \sim \mathrm{CN}(0,1)$ under the condition v_0 and $z_i \sim \mathrm{CN}(0, 1+\lambda_i)$ under the condition v_1, and z_i^2 obeys a chi-square distribution. The first term in (29) is a constant and the likelihood ratio after shifting it to the right is $\gamma_e = \ln((P_V(v_0))/(P_V(v_1))) + \sum_{i=1}^{L_w} \ln(1+\lambda_i)$. The conditional likelihood function $L'(\mathbf{z}|V)$ can be expressed as

$$\begin{cases} L'(\mathbf{z}|v_0) = \displaystyle\sum_{i=1}^{L_w} \frac{\lambda_i}{2(1+\lambda_i)} \chi^2(2) \\ L'(\mathbf{z}|v_1) = \displaystyle\sum_{i=1}^{L_w} \frac{\lambda_i}{2} \chi^2(2). \end{cases} \quad (31)$$

The conditional probability of target detection for the above conditional likelihood function at a threshold of γ_e is derived in [45] with the following expression:

$$P_{D|V,f_n,b_n}(d_1|v_0, f_n, b_n) = \sum_{i=1}^{L_w} p_i \exp\left[-\frac{\gamma_e(1+\lambda_i)}{\lambda_i} \right] \quad (32)$$

$$P_{D|V,f_n,b_n}(d_1|v_1, f_n, b_n) = \sum_{i=1}^{L_w} q_i \exp\left[-\frac{\gamma_e}{\lambda_i} \right] \quad (33)$$

where

$$\begin{cases} p_i = \displaystyle\prod_{j=1, j\neq i}^{L_w} \frac{1}{1 - \frac{\lambda_j(1+\lambda_i)}{(1+\lambda_j)\lambda_i}} \\ q_i = \displaystyle\prod_{j=1, j\neq i}^{L_w} \frac{1}{1 - \frac{\lambda_j}{\lambda_i}}. \end{cases} \quad (34)$$

So far, the specific expression of (19) has been derived.

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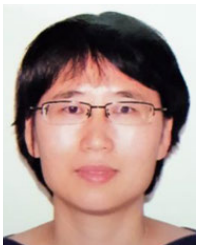
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